

Minkyung Cho

4917 BBB, 2260 Hayward Street, Ann Arbor, MI 48109

minkycho@umich.edu

<https://minkyungcho.github.io>

RESEARCH INTERESTS

Physical AI, Autonomous Driving, Robust AI, Multi-Modal, Reasoning, PEFT

EDUCATION

- **The University of Michigan** Ann Arbor, MI, USA
Ph.D. Student in Computer Science and Engineering Aug 2022 – Present
Advisor: Prof. Z. Morley Mao
- **Korea Advanced Institute of Science and Technology (KAIST)** Daejeon, Republic of Korea
M.Sc. in Computer Science Mar 2015 – Feb 2017
Advisor: Prof. Younghee Lee
- **Ewha Womans University** Seoul, Republic of Korea
B.Sc. in Computer Science and Engineering Mar 2011 – Feb 2015
Summa Cum Laude

PUBLICATIONS

1. **SiA: Synergy in Action: Benchmarking Collaborative Coordination in Multi-Arm Furniture Assembly**
Zesen Zhao*, Joseph Brewington*, **Minkyung Cho***, Ruiyang Zhu, Haoran Zhang, Hui Shen, Boyuan Zheng, Jiarui Li, Xueshen Liu, Fan Bai, Shuqing Zeng, and Z. Morley Mao
Under Submission (***Equal Contribution**)
2. **MARS: Harmonizing Multimodal Convergence via Adaptive Rank Search**
Minkyung Cho, Insu Jang, Shuowei Jin, Zesen Zhao, Adityan Jothi, Ethem F. Can, Min-Hung Chen, and Z. Morley Mao
NeurIPS 2026
3. **GraphMap: Scalable Crowd-Sourced Global Vectorized HD Map Construction via Sparse Visual Graph Fusion**
Ruiyang Zhu, **Minkyung Cho**, Shuqing Zeng, Fan Bai, and Z. Morley Mao
IROS 2026
4. **Dynamic Linear Attention**
Xin Wang, Hui Shen, Boyuan Zheng, Xueshen Liu, **Minkyung Cho**, Zhongwei Wan, Zesen Zhao, Z. Morley Mao, Mi Zhang, and Shen Yan
ICML 2026
5. **TC-LoRA: Temporally Modulated Conditional LoRA for Adaptive Diffusion Control**
Minkyung Cho, Ruben Ohana, Christian Jacobsen, Adityan Jothi, Zhuoqing Mao, Min-Hung Chen, and Ethem F. Can
NeurIPS 2025
6. **Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion**
Minkyung Cho, Yulong Cao, Jiachen Sun, Qingzhao Zhang, Marco Pavone, Jeong Joon Park, Heng Yang, and Z. Morley Mao
ICLR 2025
7. **SCORPION: Robust Spatial-Temporal Collaborative Perception Model on Lossy Wireless Network**
Ruiyang Zhu, **Minkyung Cho**, Shuqing Zeng, Fan Bai, and Z. Morley Mao
IROS 2025 (Oral)
8. **CrowdMap: Scalable Crowd-sourced Global HD Map Construction via Collaborative Map Perception and Sparse Graph Fusion**

Ruiyang Zhu*, **Minkyong Cho***, Shuqing Zeng, Fan Bai, Xiang Gao, and Z. Morley Mao
CVPRW 2025 (*Equal Contribution)

9. **Achieving the Safety and Security of the End-to-End AV Pipeline**
Noah T. Curran, **Minkyong Cho**, Ryan Feng, Liangkai Liu, Brian Jay Tang, Pedram MohajerAnsari, Alkim Domeke, Mert D. Pesé, and Kang G. Shin
ACM CCSW 2024
10. **ADoPT: LiDAR Spoofing Attack Detection based on Point-Level Temporal Consistency**
Minkyong Cho, Yulong Cao, Zixiang Zhou, and Z. Morley Mao
BMVC 2023
11. **DynaMIX: Resource Optimization for DNN-Based Real-Time Applications on a Multi-Tasking System**
Minkyong Cho and Kang G. Shin
arXiv 2023
12. **A Novel Sensitivity Metric For Mixed-Precision Quantization With Synthetic Data Generation**
Donghyun Lee*, **Minkyong Cho***, Seungwon Lee, Joonho Song, and Changkyu Choi
ICIP 2021 (*Equal Contribution)
13. **Contextual Relationship-based Activity Segmentation on an Event Stream in the IoT Environment with Multi-user Activities**
Minkyong Cho, Younggi Kim, and Younghee Lee
ACM MiddlewareW 2016
14. **Proactive Patrol Dispatch Surveillance System by Inferring Mobile Trajectories of Multiple Intruders using Binary Proximity Sensors**
Dahee Jung, **Minkyong Cho**, Omprakash Gnawali, and HyungJune Lee
IEEE INFOCOM 2016

PATENTS

1. **Apparatus and Method for Channelwise Neural Network Compression**
Wonjo Lee, Youngmin Oh, and **Minkyong Cho**
US20220114453A1. Published Apr. 14, 2022.
2. **Method for Zero-shot Pruning without Retraining**
Minkyong Cho, Searom Choi, and Seungwon Lee
US20220108180A1. Published Apr. 7, 2022.
3. **Method of replacing Bilinear Interpolation with Depthwise Transposed Convolution**
Donghyeok Kwon and **Minkyong Cho**
US20220067429A1. Published Mar. 3, 2022.
4. **A Method and An Apparatus for Performing Convolution Operations**
Songyi Han, **Minkyong Cho**, and Seungwon Lee
US20210201132A1. Published Jul. 1, 2021.
5. **Method and Apparatus for Performing Pruning of Neural Network**
Minkyong Cho, Wonjo Lee, and Seungwon Lee
US20210081798A1. Published Mar. 18, 2021. *Mounted on Samsung Galaxy S11.

ACADEMIC RESEARCH EXPERIENCE

- **Graduate Student Research Assistant (GSRA)** Ann Arbor, MI, USA
The University of Michigan (Advisor: Prof. Z. Morley Mao) Aug 2022 – Present
 - **[Vision+Language] MLLM Fine-Tuning**: Identified an imbalanced training dynamics problem and developed an MLLM fine-tuning algorithm to mitigate it. MARS accepted to NeurIPS 2026.
 - **[2D Vision+3D Vision] Multimodal Perception**: Designed a robust multimodal perception framework and proposed an uncertainty quantification and comparison algorithm across heterogeneous modalities. Cocoon accepted to ICLR 2025.

- **[Multi-Agent Visual Feature Fusion] Collaborative Perception:** Built a robust collaborative perception and HD map construction system for connected and automated vehicles. SCORPION (IROS 2025 Oral), CrowdMap (CVPRW 2025), GraphMap (IROS 2026).
 - **[3D Vision+Time] Anomaly Detection:** Developed a method for detecting anomalies by checking temporal consistency in 3D point clouds. ADoPT accepted to BMVC 2023.
- Research Intern** Ann Arbor, MI, USA
The University of Michigan (Advisor: Prof. Kang G. Shin) Feb 2021 – Aug 2022
 - **Resource Allocation:** Identified a problem in running multiple real-time vision apps on autonomous vehicle. Designed a resource optimization/allocation algorithm to satisfy apps' timing requirements.
 - **Neural Network Optimization:** Reduced resource and computational costs of NN models via mixed-precision quantization.
- Graduate Research Assistant** Daejeon, Republic of Korea
Computer Networks Lab, KAIST (Advisor: Prof. Younghee Lee) Mar 2015 – Feb 2017
 - **Activity Segmentation:** Designed and implemented an automated activity segmentation system using LSTM model.
 - **Wireless Sensor Network:** Implemented smart home/office environment using MQTT and TCP protocols, set up testbed on KAIST campus building, and managed IoT data stream from user activities.
- Undergraduate Research Assistant** Seoul, Republic of Korea
Intelligent Networked Systems Lab, Ewha Womans Univ. (Advisor: Prof. HyungJune Lee) Nov 2013 – Dec 2014
 - **Proactive Patrol Dispatch Surveillance System:** Worked on two core algorithms: 1) inferring future trajectories of multiple intruders in a building and 2) maximizing the detection probability of multiple intruders while minimizing the moving distance of the patrol officers.
 - **Wireless Sensor Network:** Implemented TinyOS-based ZigBee network consisting of TelosB motes (binary proximity sensors) and set up testbed on Ewha campus building.
- Undergraduate Research Assistant** Seoul, Republic of Korea
Security and Theory of Computing Lab, Ewha Womans Univ. (Advisor: Prof. Sang-Ho Lee) Dec 2012 – Feb 2013
 - **Visual Cryptography:** Developed joint account management algorithm in mobile banking system based on visual cryptography.

INDUSTRIAL EXPERIENCE

- Ph.D. Research Intern** Santa Clara, CA, USA
NVIDIA Research (Team: ASPIRE) May 2026 – Present
 - **VLA Reasoning:** Rethinking Groundedness for Physical Intelligence in Autonomous Driving
- Deep Learning Software Intern** Santa Clara, CA, USA
NVIDIA (Team: Metropolis) May 2025 – Aug 2025
 - **[Vision+Language+Time] Controllable Diffusion Models:** Researched optimal conditioning algorithms for generative AI models (e.g., NVIDIA Cosmos), aiming for improved conditional guidance over standard methods (e.g., ControlNet).
- Deep Learning Software Intern** Santa Clara, CA, USA
NVIDIA (Team: Metropolis) Jun 2024 – Aug 2024
 - **[Vision+Language] Efficient Fine-Tuning for Multimodal Models:** Developed parameter-efficient fine-tuning methods (e.g., LoRA) for multimodal models.
- Artificial Intelligence Researcher** Suwon, Republic of Korea
Samsung Advanced Institute of Technology @ Samsung Electronics Mar 2018 – Apr 2021
 - **Neural Network Optimization:** Designed and implemented hardware-efficient model optimization algorithms for Samsung Exynos NPU & released on Samsung AI SDK.
 - **Software/hardware Co-design:** Designed and implemented a new number system for the next-generation NPU architecture.

HONORS AND AWARDS

- **Korea National Scholarship** Mar 2015
KAIST and Korea Ministry of Science and ICT
- **Dean's List Award** Apr 2012, Oct 2012, Apr 2013, Oct 2013, Apr 2014, Oct 2014
Ewha Womans University
- **Academic Scholarship** Apr 2012, Oct 2012, Apr 2013, Oct 2013, Apr 2014, Oct 2014
Ewha Womans University
- **Han-su Scholarship** Apr 2013
Han-su Foundation

TEACHING EXPERIENCE

- **Teaching:** Main TA, Introduction to Computer Networks @ KAIST, Mar 2016 - Aug 2016
- **Counseling:** Counseling Assistant for CS Students @ KAIST, Sep 2015 - Aug 2016
- **Tutoring:** Data Structure, Operating Systems, and Java Programming @ Ewha Womans University

SERVICES

- **Tutorial Organizer:** X-Sense: Cooperative Perception and Distributed Foundation Models for Smart Mobility @ ECCV 2026, Sep 2026
- **Reading Group Organizer:** Organize Systems Reading Group @ UMich, Sep 2023 - Present
- **Conference Reviewer:** ICLR, IROS, ICML, BMVC, CPAL, NeurIPS, ICLR, CVPR